

VASCULITIS



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• Vasculitis

- *inflammation of the blood vessel wall.*
- *H. P.:*
 - Presence of peri-vascular inflammatory cell infiltrate
 - . vascular injury
 - necrosis of the endothelium,
 - fibrin deposition in blood vessel wall,
 - thrombus formation in lumen,
 - edema,
 - extravasation of erythrocytes.

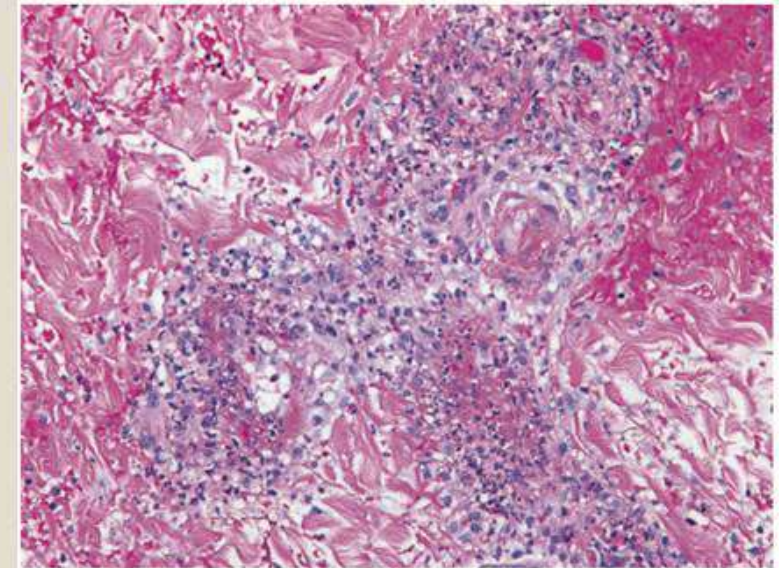


Fig. 24.4 Cutaneous small vessel vasculitis – histopathologic features. Angiocentric segmental inflammation with endothelial swelling, a neutrophilic infiltrate with leukocytoclasia, red blood cell extravasation, and fibrinoid necrosis of blood vessel walls. The neutrophils are present around and within

- **Cutaneous vasculitis may be:**

- (1) a skin-limited disease;
- (2) a primary cutaneous vasculitis with secondary systemic involvement
- (3) a cutaneous manifestation of a systemic vasculitis.

- **Vasculitis can affect small, medium-sized, or large vessels**

- . Small vessels include arterioles, capillaries, and postcapillary venules, which are found in the superficial and mid dermis of the skin.
- Medium-sized vessels refer to the small arteries and veins that reside within the deep dermis or subcutis.
- Large vessels include the aorta and named arteries.

CUTANEOUS VASCULITIS CLASSIFICATION SCHEME			
Caliber of the predominantly affected vessel	Classification	Subclassification or etiologies	Morphology of cutaneous lesions
Small	Cutaneous small vessel vasculitis (CSVV)	Henoch-Schönlein purpura, including IgA vasculitis in adults	Palpable purpura (most common)
		Acute hemorrhagic edema of infancy	Petechiae
		Urticarial vasculitis	Macular purpura
		Erythema elevatum diutinum	Urticarial papules
		Cryoglobulinemic vasculitis (types II and III)	Vesicles
		Secondary causes of CSVV (see Table 24.4):	Pustules
		- Drug exposure	Targetoid papules and plaques
		- Infections	
		- AI-CTD	
		- Malignancies, most often hematologic	
Small and medium-sized ("mixed")	ANCA-associated	Microscopic polyangiitis	Petechiae
		Granulomatosis with polyangiitis (Wegener granulomatosis)	Palpable purpura
		Eosinophilic granulomatosis with polyangiitis (Churg-Strauss syndrome)	Livedo racemosa
		Drug-induced	Retiform purpura
	Secondary causes	Infections	Ulcers
		Inflammatory disorders (e.g. AI-CTD)	Subcutaneous nodules
			Digital necrosis
Medium-sized	Polyarteritis nodosa (PAN)	Classic (systemic) PAN	Livedo racemosa
		Cutaneous PAN	Retiform purpura
			Ulcers
			Subcutaneous nodules
			Digital necrosis
Large*	Temporal arteritis		Early – erythematous or cyanotic skin, alopecia, purpura, tender nodules on frontotemporal scalp
			Late – ulceration and/or gangrene of frontotemporal scalp or tongue
	Takayasu arteritis (disease)		Erythematous subcutaneous nodules ± ulceration, pyoderma gangrenosum-like lesions on the extremities (lower > upper)
			May have evidence of small and/or medium-sized vessel vasculitis

*Cutaneous manifestations are rare.

Clinical features

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graph TD; A[Clinical features] --> B[Cutaneus]; A --> C[systemic]; B --> D[Small vessel]; B --> E[Medium];
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Cutaneus

Small vessel
Medium

systemic

Cutaneous findings



- **symptoms:**
 - usually asymptomatic, pain and pruritus.
 - Sites: mostly dependent sites (lower limbs).
 - bilateral and symmetrical.
 - Lesions: depend on the size of the blood vessels :
 -
- Small vessel vasculitis:**
- Petechiae, macular or **palpable purpura**
 - haemorrhagic vesicles, urticarial plaques.
 - E.M-like (targetoid) lesions.
 - Necrosis / ulceration³ is not usually a major feature.



◦ **Medium-sized vessel vasculitis:**

- Deep / subcutaneous nodules.
- Livedo reticularis / racemosa, retiform purpura.
- Ulcers and digital necrosis.

Systemic finding :

- constitutional symptoms as fever, weight loss and malaise.
- • Specific symptoms and signs vary according to the affected organs e.g. abdominal pain, paresthesias, hematuriam, arthralgia

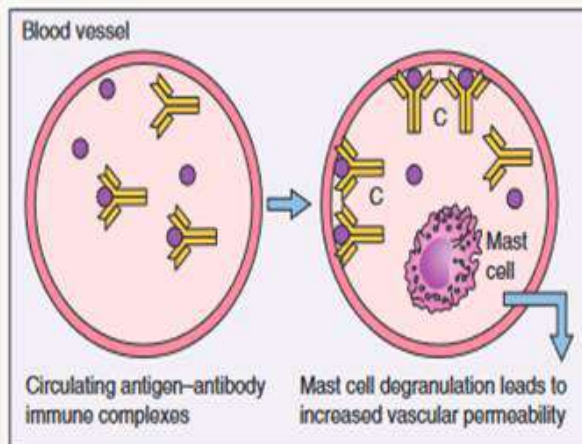


Aetiopathogenesis

PATHOGENESIS OF CUTANEOUS VASCULITIS

Immune complex-mediated

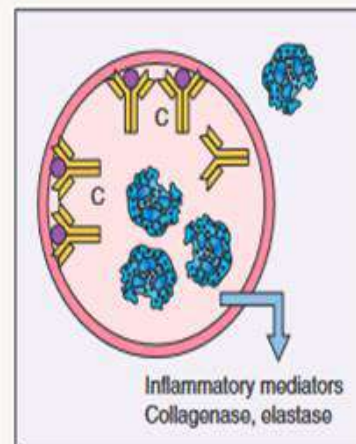
(A) Early



Immune complex deposition on blood vessel walls with subsequent activation of complement (C) leads to inflammatory cascades and mast cell recruitment

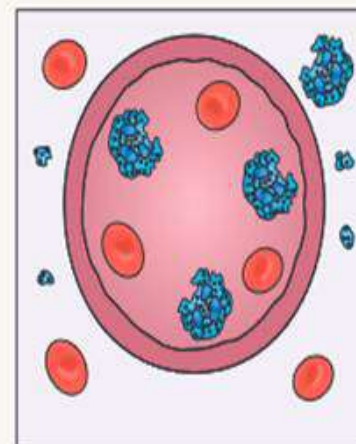


Mid



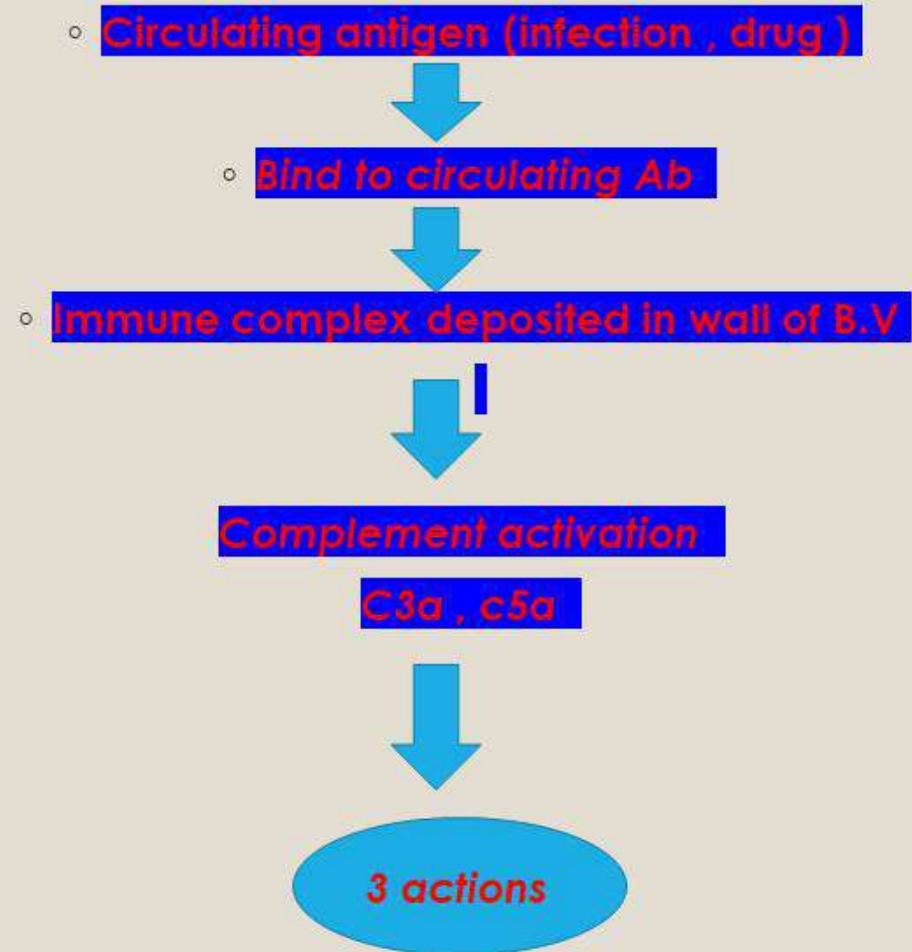
Complement-derived factors (C3a, C5a) lead to neutrophil chemotaxis. Neutrophil degranulation results in release of inflammatory mediators, collagenase and elastase

Late



Vessel wall necrosis, thrombosis, occlusion, hemorrhage

- vasculitis *is a type III hypersensitivity reaction*
- . immune complex mediated) directed against certain antigen

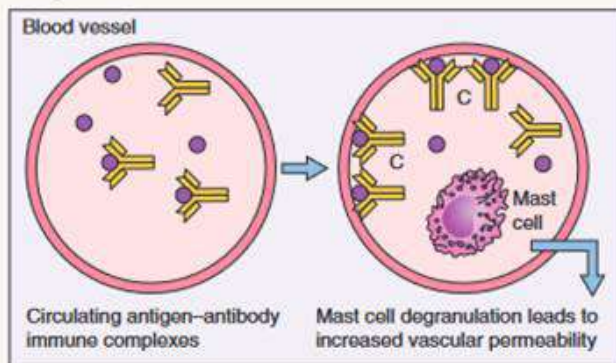


1. increase in adhesion molecules expression between inflammatory cells (**especially neutrophils**) and the endothelium (E-selectin, P-selectin, ICAM-1, VCAM-1)
2. chemotaxis of inflammatory cells mainly **neutrophils** —> degranulation —* release of inflammatory mediators and proteolytic enzymes (collagenase and elastase) —> vessel wall necrosis, thrombosis, occlusion, hemorrhage
3. **mast cell degranulation** —▶ histamine and heparin —> increased vascular dilatation and permeability —> extravasation of RBCs and perivascular inflammatory infiltrate.

PATHOGENESIS OF CUTANEOUS VASCULITIS

Immune complex-mediated

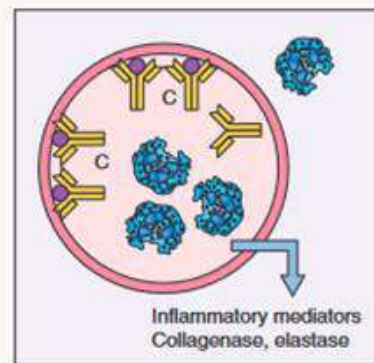
A Early



Immune complex deposition on blood vessel walls with subsequent activation of complement (C) leads to inflammatory cascades and mast cell recruitment

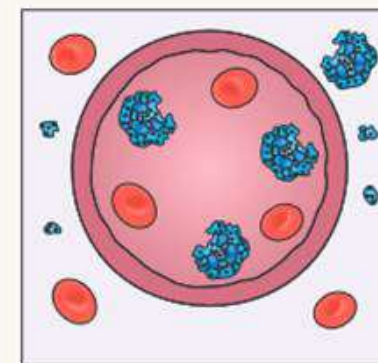


Mid



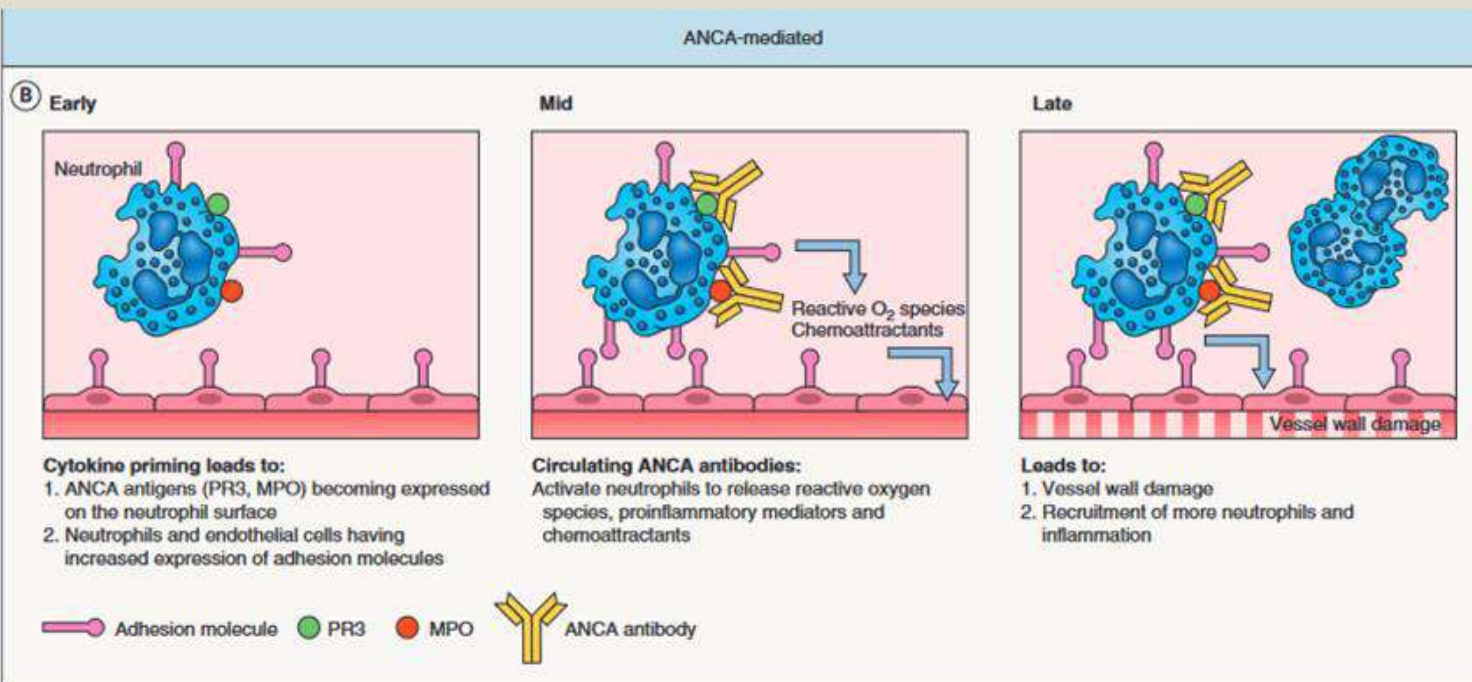
Complement-derived factors (C3a, C5a) lead to neutrophil chemotaxis
Neutrophil degranulation results in release of inflammatory mediators, collagenase and elastase

Late



◦ **ANCA (anti-neutrophil cytoplasmic antibodies):**

- Antibodies directed against neutrophil intra-cellular proteins - 2 types:
 1. Cytoplasmic ANCA (c-ANCA) against proteinase 3 (PR3)
 2. Perinuclear ANCA (p-ANCA) against myeloperoxidase (MPO)



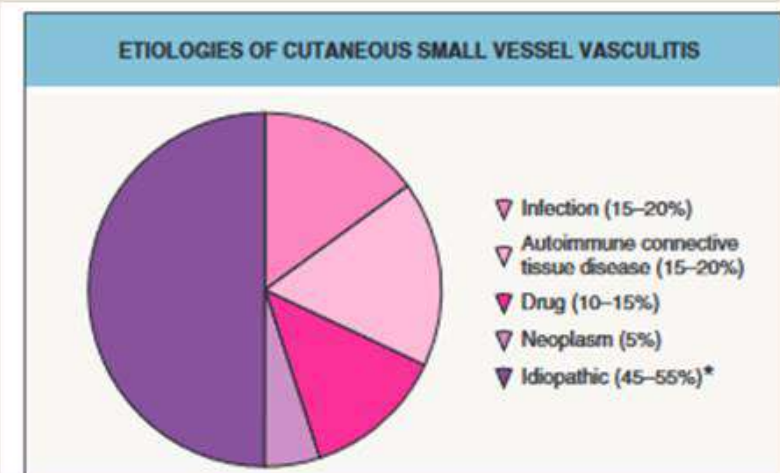
- ANCA-associated vasculitis:
- 1- Microscopic polyangiitis (MPA)
 2. Wegener's granulomatosis
 3. Churg-Strauss syndrome _

Aetiology of cutaneous vasculitis

1ry (idiopathic)
50%

- 2ry:
1. Infection
 2. Inflammation
 3. Drug
 4. Neoplasm

UNDERLYING CAUSES OF SECONDARY CUTANEOUS VASCULITIS				
Association	Incidence	Agent/Disease		
		Common	Uncommon	Rare
Idiopathic	50%			
Infection	15–20%	Bacterial		
		Beta-hemolytic streptococci, especially group A <i>Mycobacterium leprae</i>	<i>Neisseria meningococcus</i> (in chronic meningococcemia) <i>Mycobacterium tuberculosis</i> Atypical mycobacteria	<i>Mycoplasma pneumoniae</i> <i>Chlamydia pneumoniae</i> , <i>Chlamydia trachomatis</i> <i>Brucella</i> <i>Bartonella henselae</i> <i>Salmonella</i> <i>Campylobacter</i> <i>Yersinia enterocolitica</i> <i>Treponema pallidum</i>
		Septic vasculitis*		
		Infective endocarditis	<i>Neisseria meningitidis</i> (acute) <i>Neisseria gonorrhea</i> <i>Staphylococcus aureus</i> <i>Rickettsiae</i> Gram-negative rods <i>Escherichia coli</i> <i>Klebsiella</i> <i>Pseudomonas</i> Disseminated fungal infections (immunocompromised hosts) <i>Candida</i> <i>Aspergillus</i> <i>Fusarium</i> <i>Mucor</i>	<i>Francisella tularensis</i>
		Viral		
		Upper respiratory tract infection Hepatitis C > B > A, including vaccines	HIV Parvovirus B19	Cytomegalovirus Varicella zoster virus Influenza virus, including vaccine
Inflammatory disorders	15–20%	Autoimmune connective tissue diseases - Rheumatoid arthritis** - SLE - Sjögren syndrome	Inflammatory bowel disease Behçet disease Hypergammaglobulinemic purpura of Waldenström Seronegative spondyloarthropathies	Sarcoidosis Cystic fibrosis† Primary biliary cirrhosis Bowel-associated dermatosis-arthritis syndrome Gluten enteropathy



UNDERLYING CAUSES OF SECONDARY CUTANEOUS VASCULITIS

Association	Incidence	Agent/Disease		
		Common	Uncommon	Rare
Drug exposure [†]	10–15%	Antibiotics, esp. β-lactams Penicillins Cephalosporins, esp. cefaclor Sulfonamides Minocycline [†] Quinolones Macrolides Cardiovascular Thiazides Hydralazine [†] Quinidine Other Allopurinol Bortezomib D-penicillamine G-CSF NSAIDs Propylthiouracil/other antithyroid agents [†] Serum (e.g. ATG) Streptokinase	Antimicrobials Quinine Vancomycin Cardiovascular ACE inhibitors Beta-blockers Furosemide Other Cocaine adulterated with levamisole [†] COX-2 inhibitors Interferons Leukotriene inhibitors ^{†,§} Methotrexate Oral contraceptives Phenytoin Retinoids Sirolimus Sulfonylureas TNF- α inhibitors Warfarin	Antimicrobials Mefloquine Cardiovascular Amiodarone Neuropsychiatric Atypical antipsychotics Gabapentin Phenothiazines SSRIs Other Insulin Leflunomide Metformin Methamphetamine 3,4-methylenedioxymethamphetamine Rituximab SSKI Tocolytics (e.g. ritodrine, terbutaline) Miscellaneous Radiographic contrast media Food/drug additives Vitamins
Neoplasms	2–5%	Plasma cell dyscrasias - Monoclonal gammopathies - Multiple myeloma Myelodysplasia Myeloproliferative disorders Lymphoproliferative disorders Hairy cell leukemia	Solid organ carcinomas (IgA vasculitis in adults $\gg$ other forms of CSW)	
Genetic disorders	Rare	Alpha-1 antitrypsin deficiency		Immunodeficiency syndromes (see Table 60.1) Familial Mediterranean fever and other periodic fever syndromes (see Table 45.2)

1. **Infection:**

- Bacterial: Group-A Beta-hemolytic streptococci.
- Viral: Upper respiratory tract infection / Hepatitis C virus infection

2. **Systemic inflammatory disease** e.g.

- Autoimmune connective tissue diseases (SLE, RA,)
- Inflammatory bowel diseases (ulcerative colitis, Crohn's)

3. **Medication (drug intake):**

- Antibiotics: Penicillins / Sulfonamides
- NSAIDs
- Phenytoin (anti-epileptics)
- Oral contraceptives

4. **Neoplasm:** hematological (myeloproliferative /lymphoproliferative

Histeopathology

- **Level:**
 - Small vessel vasculitis: superficial and mid-dermis
 - Medium vessel vasculitis: deep dermis or subcutis (affection of small
- Findings (Criteria!):
- **- vascular injury:**
 - Endothelial cell damage (swelling, shrinkage or sloughing),
 - Fibrinoid material within the vascular lumina or walls,
 - Infiltration of the walls of by inflammatory cells,
 - vessel leakiness e.g. edema and extravasation of RBC
- **Inflammatory infiltrate:**

Varies according to the type of vasculitis (neutrophils, histiocytes ...)

 - Neutrophils undergo fragmentation of their nuclei (karyorrhexis or leukocytoclasia) leading to the production of nuclear dust.

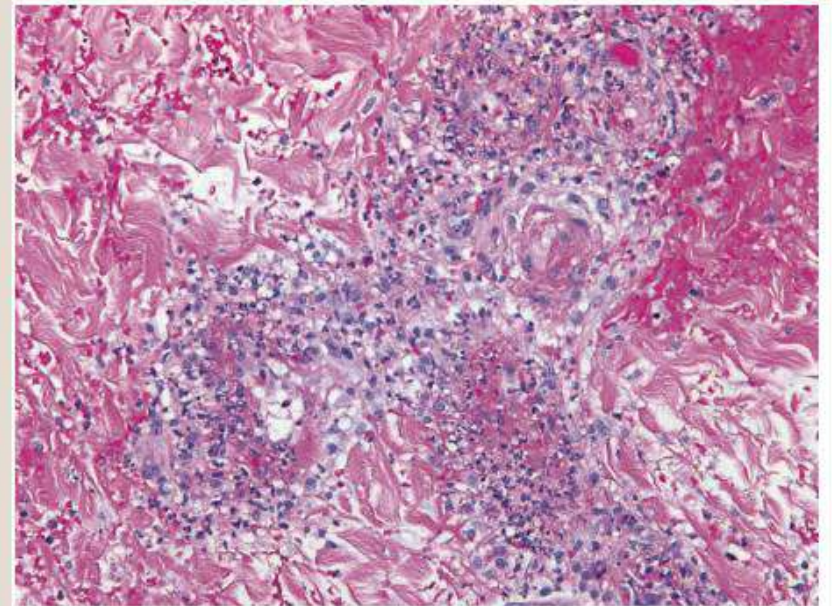


Fig. 24.4 Cutaneous small vessel vasculitis – histopathologic features. Angiocentric segmental inflammation with endothelial swelling, a neutrophilic infiltrate with leukocytoclasia, red blood cell extravasation, and fibrinoid necrosis of blood vessel walls. The neutrophils are present around and within

Classification according to size

Small vessel vasculitis

1. Csvv
2. Uv
3. EED
4. GF
5. HSP
6. AHEI
7. Cryoglobulinamia
8. ANCA

Medium

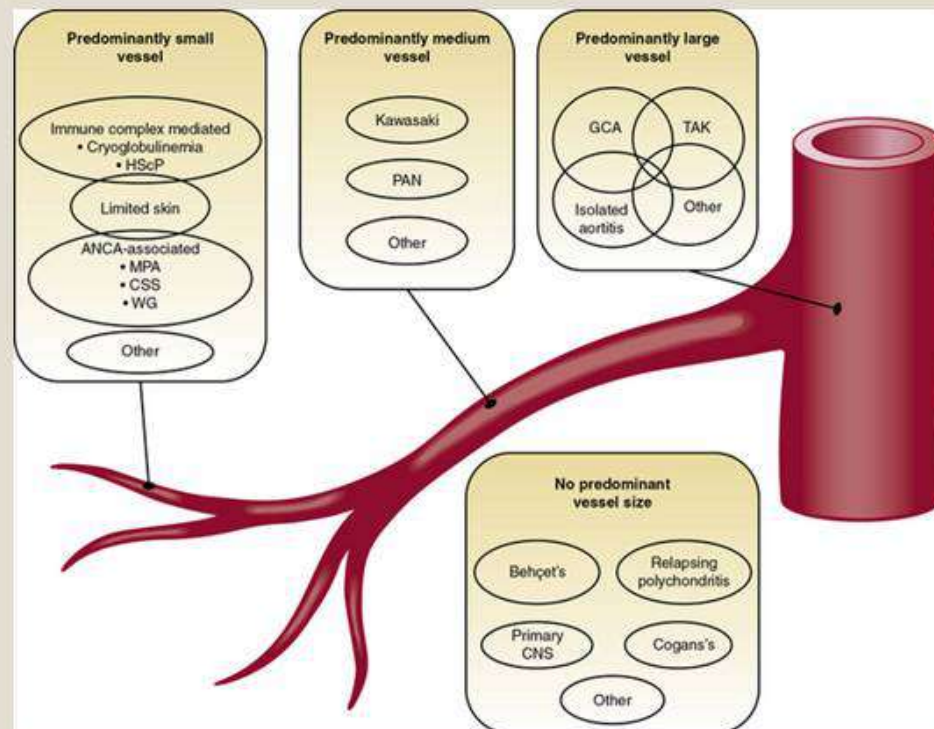
1. PAN
2. KAWASAKI

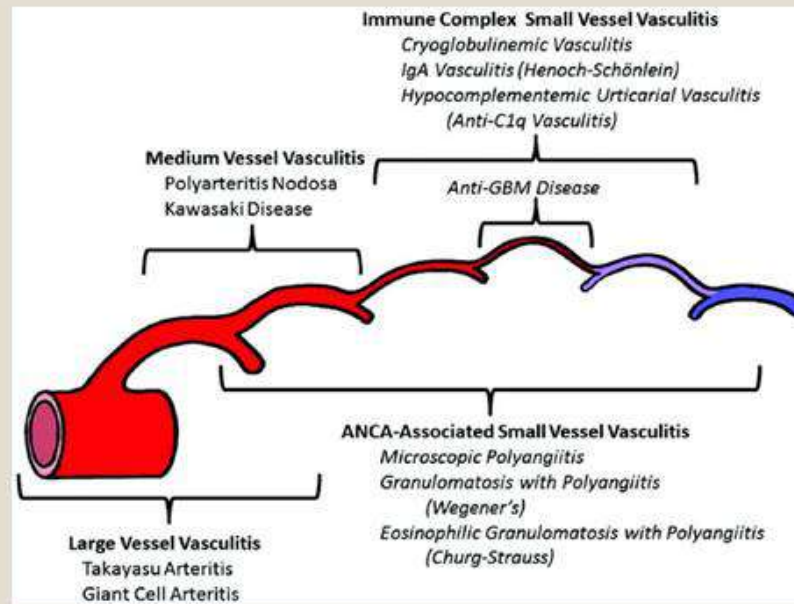
Larg size

1. Temporal arteritis
2. Takayassu arteritis

Variable vessel size :

1. Behcet syndrome
2. Others (lupus vasculitis, sarcoid vasculitis, rheumatoid vasculitis)





Classification according to infiltrate

Neutrophilic (leucocytoclastic)

1. Csxv
2. UV
3. EED
4. GF
5. HSP
6. AHEI
7. CRYOGLOBULINEMIA
8. Behcet syndrome
10. Drug-induced vasculitis
11. Lepra reaction type II (ENL)

Histiocytic (granulomatous)

1. Wegener's granulomatosis
2. Churg-Strauss syndrome
3. Giant cell (Temporal) arteritis (GCA)
4. Takayasu arteritis

Lymphocytic Eosinophilic

Classification of cutaneous vasculitis according to the 2012 Chapel Hill Consensus nomenclature:

This classification system relies on **2 main items**:

- 1. Size of the vessel affected** (small, medium, large)
- 2. Pathophysiologic mechanism**

Single-organ (skin) small-vessel vasculitis:

- Cutaneous small-vessel vasculitis (CSVV)
- Urticarial vasculitis (UV)
- Erythema elevatum diutinum (EED)
- Granuloma faciale
- Acute haemorrhagic oedema of infancy

Small-vessel immune complex-associated vasculitis:

- IgA vasculitis (Henoch-Schönlein purpura)
- Cryoglobulinaemic vasculitis

Small-vessel ANCA-associated vasculitis

- Microscopic polyangiitis (MPA)
- Wegener's granulomatosis
- Churg-Strauss syndrome

Medium-vessel vasculitis:

- Polyarteritis nodosa (PAN)
- Kawasaki disease

Large-vessel vasculitis:

- Giant cell arteritis (GCA)
- Takayasu arteritis

Cutaneous vasculitis associated with systemic disease or variable vessel size:

- Lupus vasculitis, Sarcoid vasculitis, Rheumatoid vasculitis

APPROACH TO THE PATIENT WITH SUSPECTED CUTANEOUS VASCULITIS

Patient with suspected cutaneous vasculitis

History and physical examination

- Determine if drug exposure or evidence of an underlying infection; if not, consider an associated inflammatory disorder or malignancy (see Table 24.4)
- Evaluate for extracutaneous signs and symptoms
 - Constitutional: fever, weight loss, fatigue
 - Musculoskeletal: arthralgias, myalgias
 - Renal: hematuria
 - Gastroenterologic: abdominal pain, bloody stools
 - Neurologic: numbness, paresthesias, weakness
 - Cardiopulmonary: shortness of breath, chest pain, cough, hemoptysis
 - Ear/Nose/Throat: sinusitis

Biopsy of fresh but well-developed skin lesion(s), 24–48 hours old, to confirm the presence of vasculitis and determine the size of vessels involved:

- If suspect small vessel vasculitis (e.g. if palpable purpura): punch biopsies for routine histology and direct immunofluorescence (the latter of a lesion <24 hours old)
- If suspect medium-sized vessel vasculitis: deep incisional biopsy (including subcutaneous tissue) of a nodule (preferred) or retiform purpura > the edge of an ulcer*

Pertinent positives on skin biopsy

H&E

- Granulomatous vasculitis: systemic vasculitis or systemic illness
- Lymphocytic vasculitis: AI-CTD, viral infection, drug
- Small- and medium-vessel involvement: ANCAa or AI-CTD
- Urticarial vasculitis with dermal interstitial neutrophilic infiltrate: hypocomplementemic urticarial vasculitis (consider SLE)
- HSP without eosinophils in patients >40 years old: renal involvement

DIF

- IgA predominant: HSP
- Prominent IgM: cryoglobulin or rheumatoid vasculitis
- Positive around vessels and BMZ: hypocomplementemic urticarial vasculitis (consider SLE)
- Negative: pauci-immune vasculitis (i.e. ANCA-associated)

Initial basic laboratory evaluation for all patients (repeated with flares of disease activity) to assess for extracutaneous involvement:

- CBC with differential, platelet count, ESR +/- C-reactive protein
- Hepatic panel, BUN & creatinine, urinalysis, stool guaiac

Additional evaluation for associated diseases and specific vasculitides, depending on clinical suspicion (i.e. chronic or recurrent disease, unclear cause, history and physical examination suggest underlying internal organ involvement or associated systemic disorder):

- Infection** • Antistreptolysin O and anti-DNase B titers; hepatitis B/C and HIV serologies; throat, urine or blood culture as indicated
- Inflammatory** • Cryoglobulins
- ANCA**
- AI-CTD** • Rheumatoid factor, ANA, anti-ENA antibodies (e.g. anti-Ro)
- CH50/C3/C4 if suspect urticarial vasculitis (also C1q if low C4)
- Malignancy** • Serum & urine protein electrophoresis, serum immunofixation electrophoresis, peripheral blood smear
- Age appropriate screening +/- sign/symptom directed evaluation for malignancies

Additional evaluation for extracutaneous involvement in systemic vasculitides:

ANCA associated vasculitis

- Chest X-ray, CT of chest & sinuses
- Depending on the specific condition (see Table 24.6) and clinical findings, consider: electromyogram/nerve conduction studies, echocardiogram/electrocardiogram, and biopsy of the respiratory tract (upper or lower), nerve, kidney or muscle

Clinical (extracutaneous) involvement patterns

Evaluation

- **History**
 - Triggering factor
 - systemic manifestations / complications of vasculitis
- **Examination:**
 - General examination:
 - General condition
 - Signs of systemic trigger / involvement.
- Local (Skin) examination: All the skin should be examined for:
 - Signs of cutaneous vasculitis
 - Determine extent and distribution of lesions.

Investigation :

- **Skin biopsy:**

- Should be obtained from early 'fresh' lesions (less than 48 hours old).
- **Value of skin biopsy:**
 1. Confirms the diagnosis of vasculitis
 2. Indicates size of blood vessel involvement and predominant cell type
 3. Differentiates vasculitis from thromboembolic disorders
 4. Indicates presence of granulomas
 5. Indicates certain infections e.g. mycobacteria

- **Vasculitis screen**

Treatment of cutaneous vasculitis

- Identification and removal/treatment of underlying trigger/cause
- Treatment of systemic involvement :
- Active treatment:
 - Cutaneous vasculitis frequently resolves spontaneously without treatment

- **Mild skin-limited disease:**
 - Supportive measures (e.g. leg elevation, avoiding tight clothing, rest)
 - Symptomatic therapy (e.g. antihistamines, NSAIDs)
 - Topical corticosteroids and calcineurin inhibitors
- **Chronic (>4 weeks) or more severe cutaneous disease:**
 - Colchicine (0.6 mg orally two to three times daily)
 - Dapsone (50-200 mg/day orally)
- **severe, ulcerating or progressive cutaneous disease:**
 - high-dose oral corticosteroids (up to 1 mg/kg/day prednisone)
 - **Steroid-sparing agents:**
 - **Biologic agents:**

THERAPEUTIC LADDER FOR PATIENTS WITH VASCULITIS						
Disorder	First-line treatment	Evidence levels	Second-line treatment	Evidence levels	Third-line treatment	Evidence levels
Cutaneous small vessel vasculitis	Discontinue incriminated drugs Supportive care Treat underlying infections, neoplasms NSAIDs Antihistamines	3 3	Colchicine (0.6 mg BID-TID) Dapsone (50-200 mg/day) CS	2 ⁴¹ 3 ⁴² 2 ⁴³ 3 ⁴⁴	AZA (2 mg/kg/day) MYC MTX IVIg CSA CYC Rituximab PLX	3 ⁴⁵ 3 ⁴⁶ 3 ⁴⁷ 3 ⁴⁸ 3 ⁴⁹ 3 ⁵⁰ 3 ⁵¹ 3 ⁵²
Henoch-Schönlein purpura (including IgA vasculitis in adults)	Supportive care		Dapsone Colchicine CS AZA ± CS CYC ± CS CSA ± CS Antihistamines	2 ⁵³ 3 3 1 ^{54,55} , 2 ⁵⁶ 2 ⁵⁷⁻⁵⁹ 2 ^{60,61,62} 3 ^{63,64,65}	IVIg Rituximab MYC CS + tacrolimus ACA PEX Factor XII	3 ⁶⁶ 3 ⁶⁷ 3 ^{68,69,70} 1 ⁷¹ , 2 ⁷² 3 ⁷³ 2 ⁷⁴ 2 ⁷⁵ , 3 ^{76,77} 2 ⁷⁸
Acute hemorrhagic edema of infancy	Supportive care	3 ⁷⁹			CS	3 ⁸⁰
Urticarial vasculitis	Antihistamines Indomethacin Dapsone (100-200 mg/day) ± pentoxifylline CS	3 ⁸¹ 3 ⁸² 2 ^{83,84,85} 2 ^{86,87}	Colchicine (0.6 mg BID-TID) Hydroxychloroquine (200-400 mg/day) AZA MTX	2 ^{88,89} 2 ^{90,91} 2 ^{92,93,94} 2 ^{95,96}	MYC IVIg Rituximab CSA PEX Canakinumab Tocilizumab CYC	2 ^{97,98} 3 ^{99,100} 2 ^{101,102,103} 3 ¹⁰⁴ 3 ¹⁰⁵ 1 ¹⁰⁶ 3 ¹⁰⁷ 2 ¹⁰⁸
Erythema elevatum diutinum	NSAIDs Intraleisional CS Dapsone	3 3 3	Colchicine Chloroquine Tetracyclines	3 3 3	Niacinamide PEX	3 3
Cryoglobulinemic vasculitis (+ HCV)	Peginterferon-α + ribavirin ± telaprevir or bocoprevir; sofosbuvir/ledipasvir [®] CS	1 ^{109,110} , 2 ¹¹¹ 3 ^{112,113}	CS + CYC	3 ^{114,115}	IVIg Rituximab ± ribavirin, IFN PEX	3 ¹¹⁶ 2 ^{117,118,119} 3 ¹²⁰
Cutaneous polyarteritis nodosa	Treat underlying infections Discontinue incriminated drugs NSAIDs CS (topical, intraleisional or oral)	3 2 3 3	MTX (7.5-15 mg/wk) Dapsone/sulfapyridine IVIg HCQ	3 3 3 ¹²¹ 3 ¹²²	Pentoxifylline Colchicine AZA or MYC TNF-α inhibitors Warfarin	3 3 3 3 3 ¹²³
Classic polyarteritis nodosa (+ HEV)	CS CS + PEX ± IFNα/interferon [®]	2 ¹²⁴ 2 ^{125,126,127,128}	CS + CYC	2 ^{129,130}	IVIg	3
Microscopic polyangiitis	CS CS + Rituximab ± CYC	3 ¹³¹ 1 ^{132,133}	CS + CYC MTX	3 ^{134,135} 3 ¹³⁶	AZA MYC IVIg Rituximab Infliximab	2 ^{137,138} 3 ¹³⁹ 3 ¹⁴⁰ 3 ^{141,142} 3 ¹⁴³
Granulomatosis with polyangiitis (Wegener granulomatosis) (to induce remission in limited disease)	CS + MTX	1 ^{144,145,146}	TMP-SMX (+ CS)	2 ¹⁴⁷		
Granulomatosis with polyangiitis (Wegener granulomatosis) (to induce remission in generalized disease)	CS + CYC (pulse) CS + Rituximab ± CYC CS + Rituximab	1 ^{148,149,150,151} 1 ^{152,153} , 2 ^{154,155} 1 ¹⁵⁶ , 2 ¹⁵⁷	CS + CYC + PEX (if severe renal disease, may be first line)	1 ¹⁵⁸	MYC IVIg PEX Infliximab T cell depletion Abataceptumab	3 ^{159,160} 3 ¹⁶¹ 3 3 ^{162,163} 3 3
Granulomatosis with polyangiitis (Wegener granulomatosis) (to maintain remission)	CS + AZA CS + MTX CS + Rituximab	1 ¹⁶⁴ 1 ^{165,166,167} 1 ¹⁶⁸ , 2 ¹⁶⁹	CS + CYC TMP-SMX	2 ¹⁷⁰ 1 ¹⁷¹	MYC AZA or MTX Leflunomide	2 ^{172,173,174} 3 ^{175,176} 3 ¹⁷⁷

Cutaneous small vessel vasculitis (CSVV)

Allergic cutaneous vasculitis / angiitis

- CSVV is the prototype of neutrophilic (leucocytoclastic) vasculitis
- any age / both sexes may be affected.
- **Differential diagnosis :**
 - **Purpuric macules or patch**
 - Arthropod bites
 - Morbilliform drug eruptions with hemorrhage
 - Erythema multiforme
 - Pityriasis lichenoides et varioliformis acuta (PLEVA)
 - Infectious emboli (septic vasculitis)
 - Pigmented purpuric dermatose

Diagnostic criteria

○ The American College of Rheumatology (ACR, 1990)

The presence of three of the following five criteria:

1. Age > 16 years at disease onset
2. History of taking a medication at onset (a precipitating factor)
3. The presence of palpable purpura
4. The presence of a maculopapular rash
5. A biopsy demonstrating granulocytes around an arteriole or venule



Fig. 24.6 Cutaneous small vessel vasculitis associated with systemic disorders. The underlying diseases were Sjögren syndrome (A), rheumatoid arthritis (B), relapsing polychondritis with myelodysplasia (C), and endocarditis due to *Streptococcus mitis* (D). Note the Koebner phenomenon in A and rheumatoid nodules in B. C, Courtesy, Jean I. Bologna, MD.



Fig. 24.3 Clinical variants of cutaneous small vessel vasculitis. **A** Targetoid appearance that can resemble erythema multiforme. **B** Hemorrhagic vesicles in addition to palpable purpura. **C** Hemorrhagic crusts in an annular configuration. **D** Lesions limited to the upper extremities. **E** Purpuric macules and papules with areas of confluence that resembles purpuric morbilliform eruption.

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2-Urticarial vasculitis (UV)

- UV is a clinic-pathological entity characterized:
 1. clinically by chronic recurrent episodes of painful persistent urticarial lesions (wheals) .
 2. pathologically by leucocytoclastic vasculitis. ____
- more common in females 30-50 yrs
- . SLE and Sjogren syndrome
 - Viral infections e.g. hepatitis (HBV, HCV) and Epstein-Barr (EBV)



Classification

Hypo-complementaemic UV (HUV)	Normo-complementaemic UV
Less common (20-30 %)	More common (70-80%)
Low serum complement • Autoantibodies against C1q (100%) • Decreased function of C1q	Normal serum complement
More severe	More benign
Constitutional symptoms and systemic affection may be present	Skin-limited disease
Overlap with SLE • 35% of SLE patients have anti-C1q • 25% of H UV have anti-dsDNA	NO overlap with SLE

So, UV is better considered as a spectrum:

1. Normo-complementaemic UV: skin lesions only.
2. Hypo-complementaemic UV: skin lesions with hypocomplementaemia.
3. UV with SLE criteria SLE.

Clinical picture

- **4 P**

- **Persistent, painful, purpura, pih, non blanchable**
- Trunk, proximal extremities
- Chronic recurrent wheals
- Constitutional symptoms

Systemic manifestations :with HUV.

- Cardiopulmonary: cough, dyspnoea,
- Renal: proteinuria or haematuria
- Gastrointestinal: abdominal discomfort,
- Arthritis
- Ocular: episcleritis, uveitis, conjunctivitis
- Neurological: meningitis, nerve palsies and transverse myelitis



- Differential diagnosis
- (WHEAL LIKE) STEM
 1. Urticaria
 2. Urticarial phase of bullous pemphigoid
 3. Early inflammatory morphea
 4. Tumid LE, Sweet syndrome, Erythema multiforme
- Therapeutic ladder: dapsone, colchicine, hydroxychloroquine, prednisolone, immunosuppressive agents

- **HUV syndrome**

- Two major criteria: (1) urticaria for 6 months and (2) hypocomplementemia
- plus two or more minor criteria: **VAUG**
 - (1) vasculitis
 - (2) arthralgia or arthritis;
 - (3) uveitis
 - (4) glomerulonephritis
 - (5) recurrent abdominal pain;

3-Erythem a elevatumdiutinum(EED)

- rare chronic small-vessel vasculitis.
- **Aetiology:**
 - (HIV) infection, strept
 - SLE,
- **c/p :**
 - red -brown or yellowish papules, plaques or nodules
 - symmetrically distributed – acral and peri-articular sites
 - Early: the lesions are erythematous and soft
 - Late: lesions are firm (fibrosis) leaving atrophic scars.



- With arthritis , eye affection
- H/P:
 - EARLY :LCV +EOSINOPHILS
 - Late:fibrosis
- Dd
- Early sweet , DH
- Late: **GHERX(PAPULES ON JOINT)**



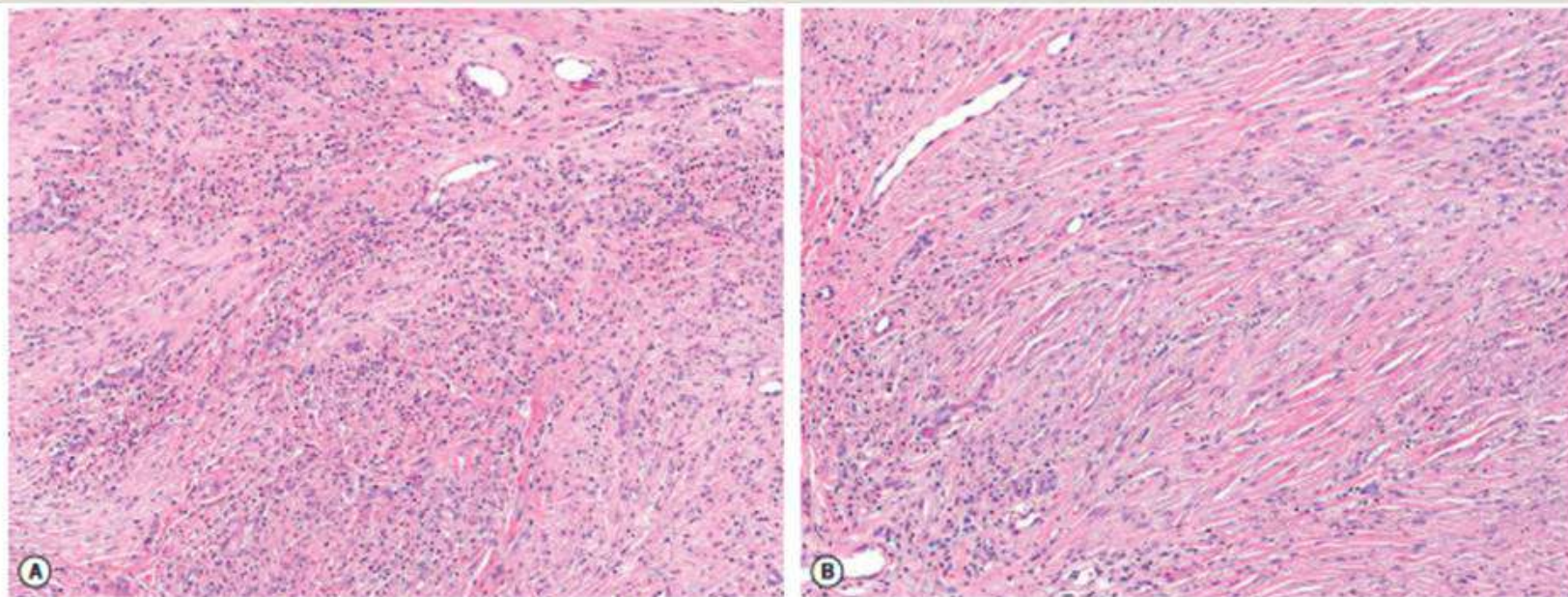


Fig. 24.12 Erythema elevatum diutinum – histologic features. **A** An early-stage lesion: dense neutrophilic infiltrates with focal leukocytoclastic vasculitis. **B** A late-stage lesion: a sparser inflammatory infiltrate and marked fibrosis. *Courtesy, Lawrence E Gibson, MD.*

- Treatment :

- 1st line :DAPSON

- 2nd line : topical steroid , ILC

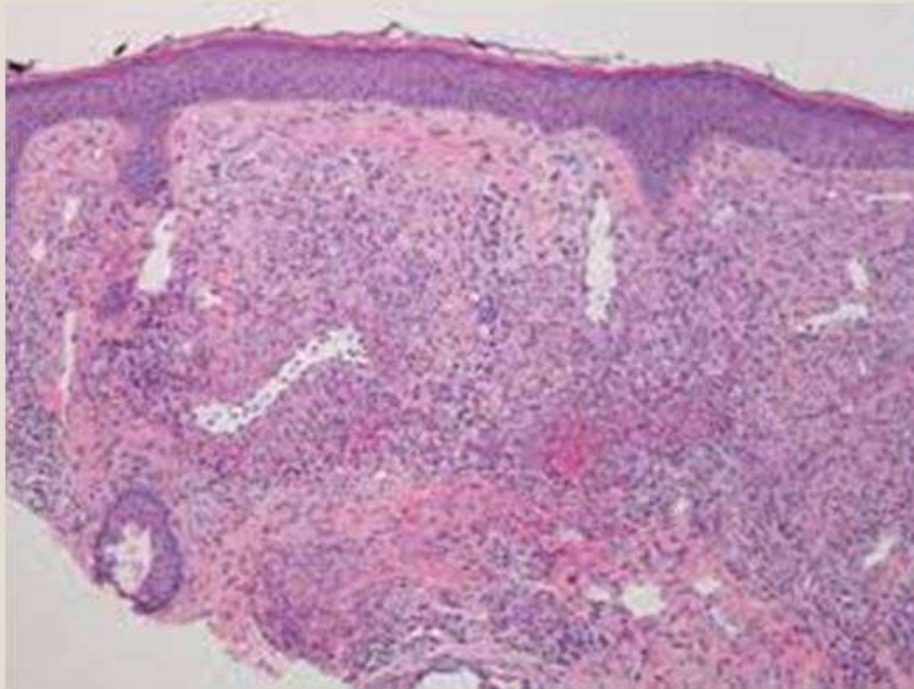
4-Granulom a faciale (facial granuloma with eosinophilia)

- adults (40 - 60 years old).
more common in males
- Solitary plaque on face with prominent follicular opening
- Limited to skin no any systemic
- H/P
 - LCV , eosinophilis ,
 - GRENZ ZONE :band of normal collagen separates the inflammatory infiltrate from the overlying epidermis
- RESIST to treatment
 - 1st line :tarulimus
 - 2nd line :topical , intralesional steroid





- DD
 - Lymphoma and pseudolymphoma,
 - sarcoidosis,
 - tumid lupus erythematosus, fixed drug eruption, rhinophyma, leprosy, TB



- DD
 - LI
 - Lymphocytoma cutis
 - B cell lymphoma
 - DF

5-Acute haemorrhagic oedema of infancy (A H E I)

- AHEI is a benign, cutaneous, leucocytoclastic vasculitis
- Finkelstein-Seidlmayer disease
 - Purpura en cocarde avec edema usually affects children < 2 years more common in boys (-70%). winter months
- 75% of patients have an associated infection, drug exposure or immunization



- large erythematous patches or urticarial plaques with annular, targetoid purpuric plaques.
- Vesicles, bullae
non-pitting edema
- face (cheeks, ears), extremities
- Constitutional symptoms.
- Systemic affection: rare. Oral petechiae,
- conjunctival injection, abdominal pain, arthralgias, glomerulonephritis.



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TRIGGERS AND ASSOCIATIONS IN ACUTE HEMORRHAGIC EDEMA OF INFANCY AND URTICARIAL VASCULITIS

Acute hemorrhagic edema of infancy – triggers

Infections

- Adenovirus
- Coxsackievirus
- Cytomegalovirus
- Epstein-Barr virus
- Herpes simplex virus
- Hepatitis A virus
- Measles
- Rotavirus
- Varicella zoster virus
- Campylobacter spp.
- Escherichia coli
- Mycobacterium tuberculosis
- Streptococcus spp.

Medications

- Acetaminophen
- Penicillins
- Cephalosporins
- Trimethoprim-sulfamethoxazole
- NSAIDs

Urticarial vasculitis – associations

Autoimmune connective tissue diseases (Sjögren syndrome, SLE)

Serum sickness

Cryoglobulinemia

Infections

- Hepatitis B virus
- Hepatitis C virus
- Epstein-Barr virus
- Lyme disease

Medications

- Cimetidine
- Cocaine
- Diltiazem
- Elinorecept
- Fluoxetine
- Infliximab
- Methotrexate
- NSAIDs
- Potassium iodide
- Procainamide
- Procarbazine

Hematologic malignancies

- Plasma cell dyscrasias (IgM, IgG, IgA)
- Leukemias
- Lymphomas
- Castleman disease

Solid organ malignancies – rare

- Colon carcinoma
- Renal cell carcinoma

- DIF, IgA deposits in a peri-vascular pattern are seen in % to % of cases
- Supportive treatment (antibiotics for concurrent bacterial infections
anti-histamines .
- Systemic corticosteroids DO NOT alter the course o f the disease

	AHEI	HSP
Age	< 2 years	< 10 years
Triggers	Infections (viral), drugs	Infection (Strept.)
Purpura	Cockade pattern	Petechiae
Edema	Constant	Less evident
Systemic affection	Rare	Common
IgA (DIF)	¼ - ⅓ of cases	100%
Resolution	More rapid	Slower
Relapse	Rare	May occur

Henoch-Schdnlein purpura(HSP)

IgA vasculitis

- Anaphylactoid purpura

- Small vessel vasculitis characterized by **IgA dominant** immune deposits affecting skin, gastrointestinal tract, joints and kidneys.
- HSP is the most common form of vasculitis in children.
- **Streptococcal upper respiratory tract infections**
- Immune complex mediated (type III hypersensitivity)



- **Tetrad**
- **Skin** : palpabl purpra ,
 - Buttouks ,extremeties
 - Bilateral symmetrical
- **Arthritis (75%)**: painful - most frequently affecting knees and ankles.
- **Gastrointestinal involvement (50-75%)**: colicky abdominal pain, gastrointestinal bleeding
- **Renal involvement (40-50%)**: microscopic or gross hematuria and proteinuria.

DD

AHEI.

- Others: idiopathic thrombocytopenic purpura

Diagnostic criteria

The ACR classification criteria (1990):

Presence of any two of the following four criteria:

1. Palpable purpura
2. Bowel angina
3. Age < 20 years at onset
4. The presence of granulocytes in the vessel wall on biopsy

Modified ACR CRITERIA

- Palpable purpura (must be present) plus at least one of the following:
 1. arthritis or arthralgia
 2. diffuse abdominal pain
 3. renal involvement (hematuria and/or proteinuria)
 4. any biopsy demonstrating predominant IgA deposition
- DIF: perivascular IgA, C3, fibrin deposition

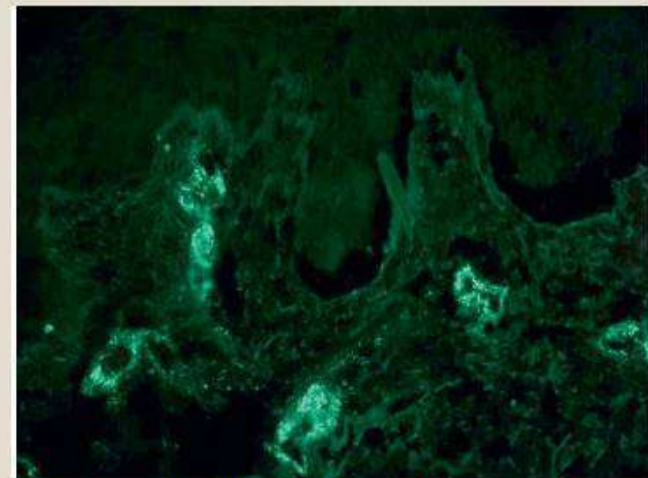


Fig. 24.8 Direct immunofluorescence microscopy of Henoch-Schönlein purpura. Strong staining for IgA within the walls of dermal blood vessels. Used with permission from Poterucha TJ, et al. *J Am Acad Dermatol.* 2012;67:612-6.

- Treatment :

- Skin lesions:

- supportive treatment (HSP is self limited and resolves spontaneously).

- Dapsone, colchicines, steroids, IVIg, Rituximab

-

- systemic affection

- systemic steroids (prednisolone 1 mg/kg/day) for 2 weeks

- tapering over 2 weeks.



CoNNect Academy - Derma - By Dr. Eman Gomaa